

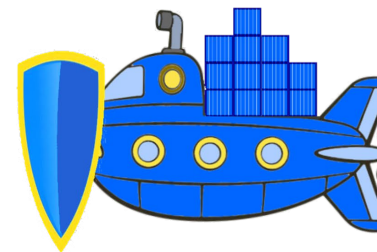
T5.3

Smart Media Distribution validation

UPM



POLITÉCNICA



CyberNEMO



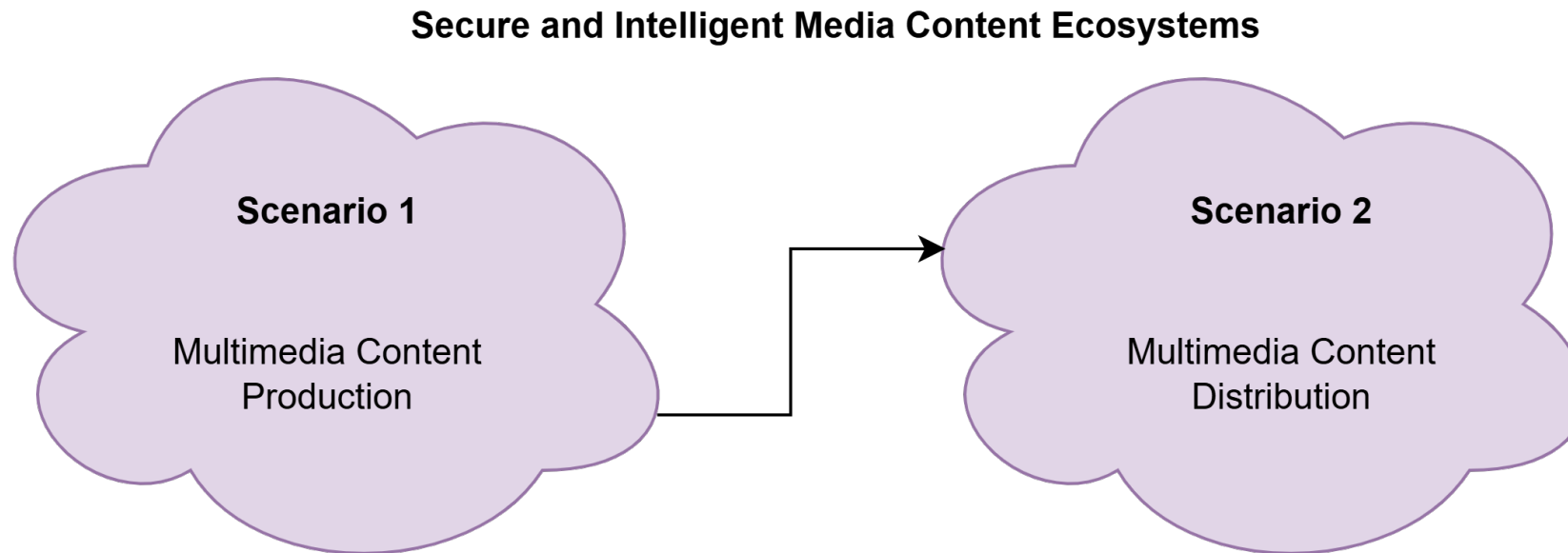
T5.3: Smart Media Distribution validation

- **Leader**
 - UPM
- **Contributors**
 - UC3M, DNV, MAG
- **Duration**
 - M9-M36
- **Objective**
 - Evaluate how CyberNEMO solutions (ZTNA and ZT-FML), can identify vulnerabilities and prevent attacks in IOT, Edge and Cloud environments
 - Ensure the correct functioning of the system by integrating with European cybersecurity platforms and standards, measuring its performance in threat detection, incident response and data protection



T5.3: Smart Media Distribution validation

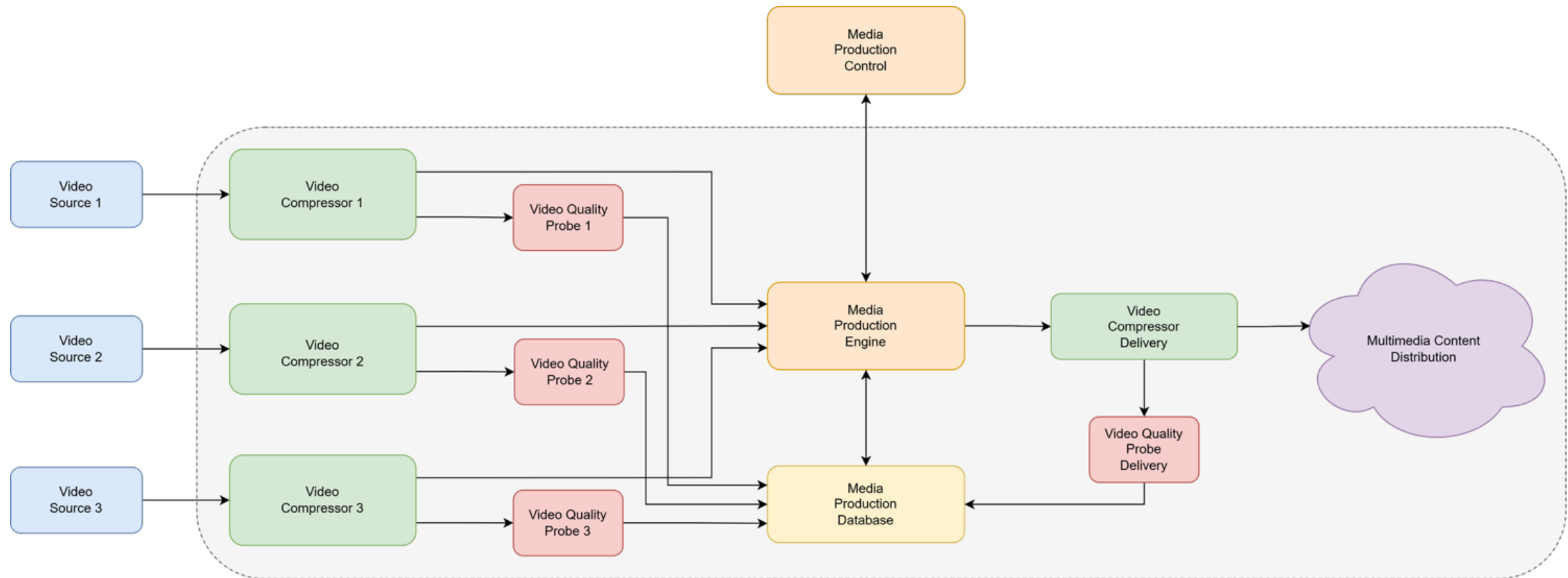
- **UC1: Collaborative and Secure Multimedia Content Production**
- **UC2: Efficient and Secure Multimedia Content Distribution**





T5.3: Smart Media Distribution validation

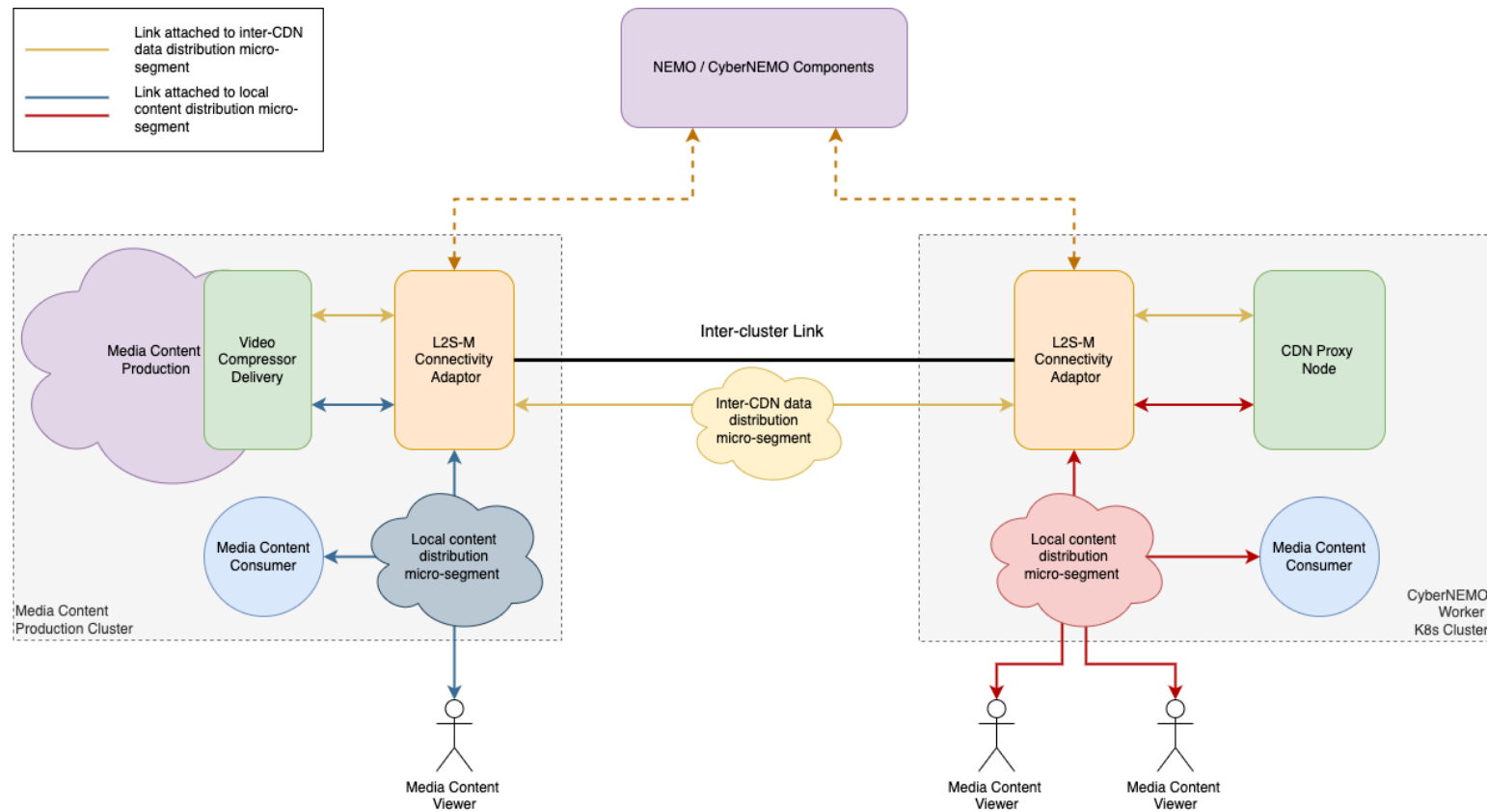
- **UC1: Collaborative and Secure Multimedia Content Production (UPM)**

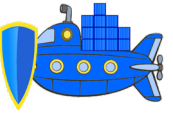




T5.3: Smart Media Distribution validation

- **UC2: Efficient and Secure Multimedia Content Distribution (UC3M)**





T5.3: Smart Media Distribution validation

Status updates since the last plenary meeting

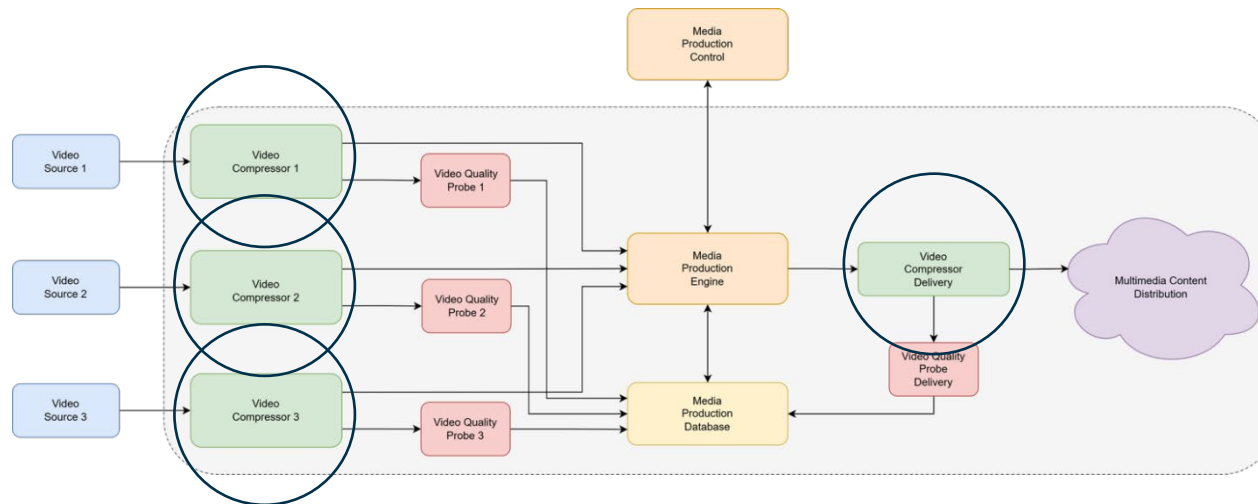
- Trial components deployment in the UPM local cluster ([ongoing work](#))
- Integration and connection of the local cluster with the CyberNEMO cluster ([next steps](#))
- Testing and validation of CyberNEMO security components ([next steps](#))



T5.3: Smart Media Distribution validation

Ongoing work

- Trial components deployment. Video Compressor



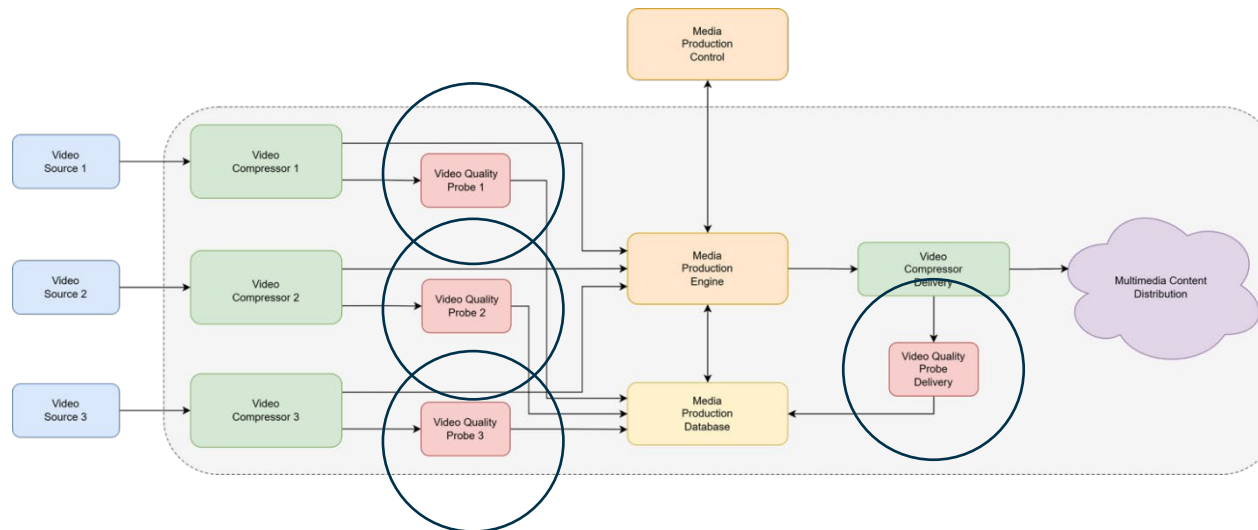
```
arya@arya-TEKNOPRO: ~/Desktop/Cybernemo/vcompressor
2026-04-09 12:41:21 arya-TEKNOPRO __main__[1] INFO {'type': 'vcompressor', 'input': 'tcp://192.168.0.42:15000', 'output': 'udp://localhost:5678', 'codec_name': 'libx264', 'resolution': '1920x1080', 'frame_rate': '50', 'bitrate_video': '4000k', 'preset': 'medium', 'bframes': '3', 'gop_size': '50', 'audio_codec_name': 'aac', 'bitrate_audio': '128k', 'sample_rate': '48000', 'channels': '2', 'timestamp': '2026-04-09T12:41:21.646125', 'ffmpeg_speed': 1.0, 'ffmpeg_fps': 25.0, 'ffmpeg_frame': 12220, 'ffmpeg_q': 20.0, 'ffmpeg_bitrate_kbps': 4216.4, 'ffmpeg_dropped_frames': 0, 'process_cpu_percent': 200.9, 'process_memory_mb': 835.87, 'process_virtual_memory_mb': 2200.98, 'process_threads': 47}
2026-04-09 12:41:26 arya-TEKNOPRO __main__[1] INFO {'type': 'vcompressor', 'input': 'tcp://192.168.0.42:15000', 'output': 'udp://localhost:5678', 'codec_name': 'libx264', 'resolution': '1920x1080', 'frame_rate': '50', 'bitrate_video': '4000k', 'preset': 'medium', 'bframes': '3', 'gop_size': '50', 'audio_codec_name': 'aac', 'bitrate_audio': '128k', 'sample_rate': '48000', 'channels': '2', 'timestamp': '2026-04-09T12:41:26.845986', 'ffmpeg_speed': 1.0, 'ffmpeg_fps': 25.0, 'ffmpeg_frame': 12350, 'ffmpeg_q': 20.0, 'ffmpeg_bitrate_kbps': 4212.8, 'ffmpeg_dropped_frames': 0, 'process_cpu_percent': 229.0, 'process_memory_mb': 835.88, 'process_virtual_memory_mb': 2200.98, 'process_threads': 47}
2026-04-09 12:41:32 arya-TEKNOPRO __main__[1] INFO {'type': 'vcompressor', 'input': 'tcp://192.168.0.42:15000', 'output': 'udp://localhost:5678', 'codec_name': 'libx264', 'resolution': '1920x1080', 'frame_rate': '50', 'bitrate_video': '4000k', 'preset': 'medium', 'bframes': '3', 'gop_size': '50', 'audio_codec_name': 'aac', 'bitrate_audio': '128k', 'sample_rate': '48000', 'channels': '2', 'timestamp': '2026-04-09T12:41:32.044808', 'ffmpeg_speed': 1.0, 'ffmpeg_fps': 25.0, 'ffmpeg_frame': 12480, 'ffmpeg_q': 20.0, 'ffmpeg_bitrate_kbps': 4215.6, 'ffmpeg_dropped_frames': 0, 'process_cpu_percent': 228.9, 'process_memory_mb': 835.89, 'process_virtual_memory_mb': 2200.98, 'process_threads': 47}
2026-04-09 12:41:37 arya-TEKNOPRO __main__[1] INFO {'type': 'vcompressor', 'input': 'tcp://192.168.0.42:15000', 'output': 'udp://localhost:5678', 'codec_name': 'libx264', 'resolution': '1920x1080', 'frame_rate': '50', 'bitrate_video': '4000k', 'preset': 'medium', 'bframes': '3', 'gop_size': '50', 'audio_codec_name': 'aac', 'bitrate_audio': '128k', 'sample_rate': '48000', 'channels': '2', 'timestamp': '2026-04-09T12:41:37.205664', 'ffmpeg_speed': 1.0, 'ffmpeg_fps': 25.0, 'ffmpeg_frame': 12609, 'ffmpeg_q': 20.0, 'ffmpeg_bitrate_kbps': 4212.1, 'ffmpeg_dropped_frames': 0, 'process_cpu_percent': 201.5, 'process_memory_mb': 835.89, 'process_virtual_memory_mb': 2200.98, 'process_threads': 47}
2026-04-09 12:41:42 arya-TEKNOPRO __main__[1] INFO {'type': 'vcompressor', 'input': 'tcp://192.168.0.42:15000', 'output': 'udp://localhost:5678', 'codec_name': 'libx264', 'resolution': '1920x1080', 'frame_rate': '50', 'bitrate_video': '4000k', 'preset': 'medium', 'bframes': '3', 'gop_size': '50', 'audio_codec_name': 'aac', 'bitrate_audio': '128k', 'sample_rate': '48000', 'channels': '2', 'timestamp': '2026-04-09T12:41:42.406489', 'ffmpeg_speed': 1.0, 'ffmpeg_fps': 25.0, 'ffmpeg_frame': 12739, 'ffmpeg_q': 19.0, 'ffmpeg_bitrate_kbps': 4211.7, 'ffmpeg_dropped_frames': 0, 'process_cpu_percent': 210.2, 'process_memory_mb': 835.89, 'process_virtual_memory_mb': 2200.98, 'process_threads': 47}
2026-04-09 12:41:47 arya-TEKNOPRO __main__[1] INFO {'type': 'vcompressor', 'input': 'tcp://192.168.0.42:15000', 'output': 'udp://localhost:5678', 'codec_name': 'libx264', 'resolution': '1920x1080', 'frame_rate': '50', 'bitrate_video': '4000k', 'preset': 'medium', 'bframes': '3', 'gop_size': '50', 'audio_codec_name': 'aac', 'bitrate_audio': '128k', 'sample_rate': '48000', 'channels': '2', 'timestamp': '2026-04-09T12:41:47.606076', 'ffmpeg_speed': 1.0, 'ffmpeg_fps': 25.0, 'ffmpeg_frame': 12869, 'ffmpeg_q': 20.0, 'ffmpeg_bitrate_kbps': 4220.3, 'ffmpeg_dropped_frames': 0, 'process_cpu_percent': 223.7, 'process_memory_mb': 835.9, 'process_virtual_memory_mb': 2200.98, 'process_threads': 47}
```



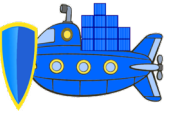

T5.3: Smart Media Distribution validation

Ongoing work

- Trial components deployment. Video Quality Probe



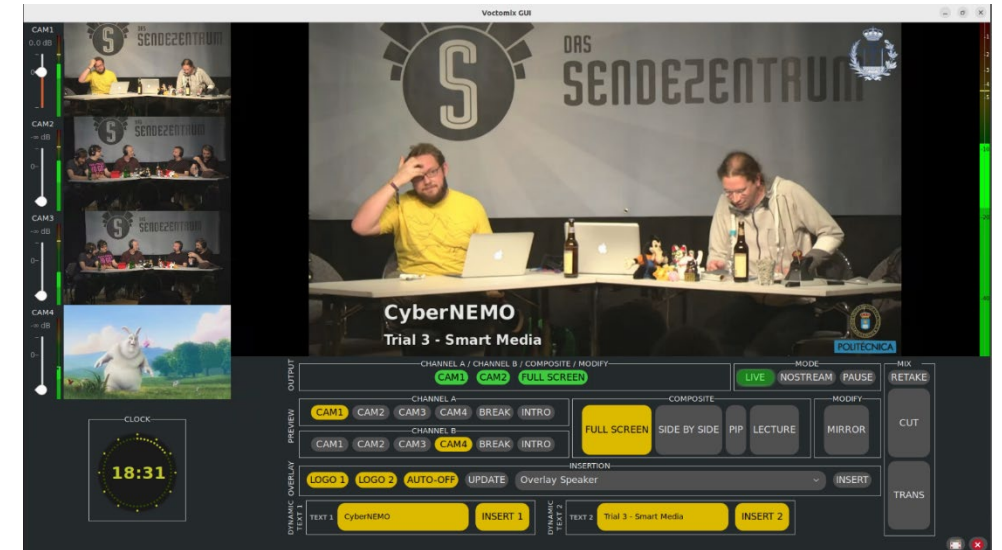
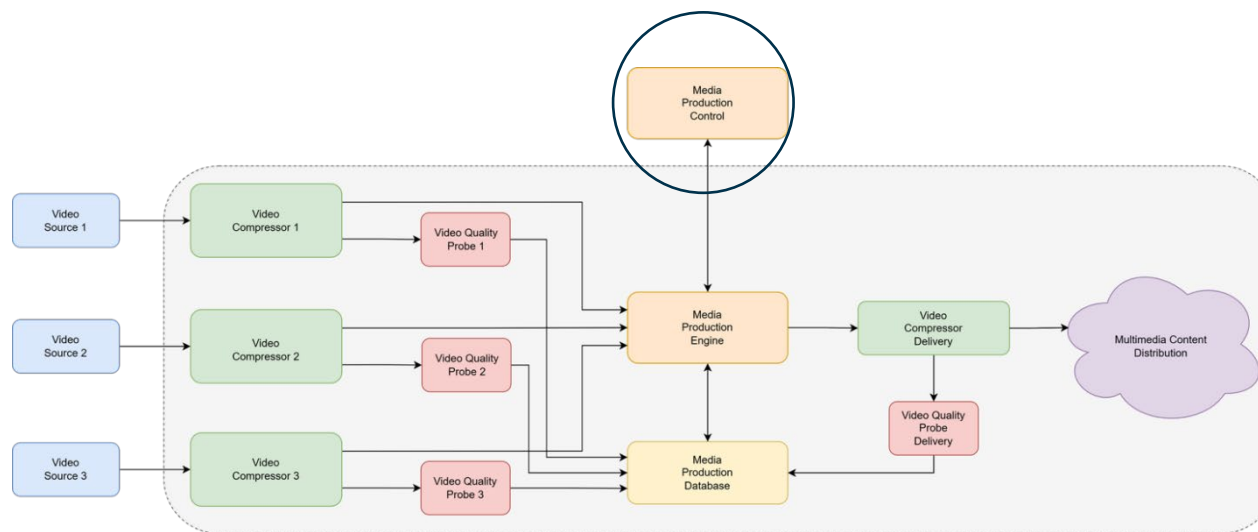
```
arya@arya-TEKNOPRO: ~/Desktop/Cybernemo/vqprobe
2026-04-09 12:42:06 arya-TEKNOPRO __main__[1] INFO {'type': 'vqprobe', 'path': 'udp://localhost:5678', 't
imestamp': '2026-04-09T12:42:06.358219', 'width': 1920, 'height': 1080, 'frame_rate': 50, 'num_frames': 1
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nnels': 2, 'codec_name': 'h264', 'profile': 'High', 'level': 42, 'pix_fmt': 'yuv420p', 'color_space': 'bt
709', 'color_transfer': 'bt709', 'color_primaries': 'bt709', 'color_range': 'tv', 'field_order': 'progres
sive', 'has_b_frames': 2, 'refs': 1, 'queue_size': 2, 'alert_total_dropped_segments': 17, 'process_cpu_pe
rcent': 53.5, 'process_memory_mb': 5525.23, 'process_threads': 23, 'spatialactivity': 53.23377613333336,
'temporalactivity': 6.132893892617449, 'blur': 3.55536, 'blockiness': 0.907786933333335, 'blockloss': 2
.435714333333333, 'freezing': 0.0, 'letterbox': 0.0, 'pillarbox': 0.0, 'blackout': 0.0, 'exposure': 133.7
4, 'contrast': 49.30835673333333, 'interlace': 0.000665, 'noise': 0.3500000000000001, 'flickering': 0.621
485555555556, 'volume_mean': -18.7, 'volume_max': -4.0, 'dropped_frames': 0, 'alert_dropped_frames': Fal
se, 'alert_blur': False, 'alert_blockiness': False, 'alert_blockloss': False, 'alert_freezing': False, 'a
lert_uniform_frame': False, 'alert_black_frame': False, 'alert_no_audio': False, 'alert_silence': False,
'count_alert_dropped_frames': 0, 'count_alert_blur': 1, 'count_alert_blockiness': 0, 'count_alert_blocklo
ss': 0, 'count_alert_freezing': 0, 'count_alert_uniform_frame': 0, 'count_alert_black_frame': 0, 'count_a
lert_no_audio': 0, 'count_alert_silence': 0, 'count_measurements': 86, 'mos_value': 3.33}
2026-04-09 12:42:11 arya-TEKNOPRO __main__[1] INFO {'type': 'vqprobe', 'path': 'udp://localhost:5678', 't
imestamp': '2026-04-09T12:42:11.332250', 'width': 1920, 'height': 1080, 'frame_rate': 50, 'num_frames': 1
00, 'duration': '4.076000', 'bitrate': '4240054', 'audio_codec_name': 'aac', 'sample_rate': '48000', 'cha
nnels': 2, 'codec_name': 'h264', 'profile': 'High', 'level': 42, 'pix_fmt': 'yuv420p', 'color_space': 'bt
709', 'color_transfer': 'bt709', 'color_primaries': 'bt709', 'color_range': 'tv', 'field_order': 'progres
sive', 'has_b_frames': 2, 'refs': 1, 'queue_size': 2, 'alert_total_dropped_segments': 17, 'process_cpu_pe
rcent': 56.1, 'process_memory_mb': 5544.79, 'process_threads': 23, 'spatialactivity': 51.14091989999999,
'temporalactivity': 5.56218, 'blur': 3.6712700000000003, 'blockiness': 0.9280957, 'blockloss': 2.1900003,
'freezing': 0.0, 'letterbox': 0.0, 'pillarbox': 0.0, 'blackout': 0.0, 'exposure': 133.61, 'contrast': 49
.064432400000001, 'interlace': 0.0006691, 'noise': 0.3500000000000001, 'flickering': 0.6675141666666667, '
volume_mean': -18.5, 'volume_max': -3.9, 'dropped_frames': 0, 'alert_dropped_frames': False, 'alert_blur'
: False, 'alert_blockiness': False, 'alert_blockloss': False, 'alert_freezing': False, 'alert_uniform_fra
me': False, 'alert_black_frame': False, 'alert_no_audio': False, 'alert_silence': False, 'count_alert_dro
pped_frames': 0, 'count_alert_blur': 1, 'count_alert_blockiness': 0, 'count_alert_blockloss': 0, 'count_a
lert_freezing': 0, 'count_alert_uniform_frame': 0, 'count_alert_black_frame': 0, 'count_alert_no_audio':
0, 'count_alert_silence': 0, 'count_measurements': 87, 'mos_value': 3.33}
2026-04-09 12:42:11 arya-TEKNOPRO __main__[1] WARNING Delay detected. Queue is: 3. Deleting 1 segments.
2026-04-09 12:42:16 arya-TEKNOPRO __main__[1] INFO {'type': 'vqprobe', 'path': 'udp://localhost:5678', 't
imestamp': '2026-04-09T12:42:16.316032', 'width': 1920, 'height': 1080, 'frame_rate': 50, 'num_frames': 1
00, 'duration': '4.070667', 'bitrate': '4816445', 'audio_codec_name': 'aac', 'sample_rate': '48000', 'cha
nnels': 2, 'codec_name': 'h264', 'profile': 'High', 'level': 42, 'pix_fmt': 'yuv420p', 'color_space': 'bt
709', 'color_transfer': 'bt709', 'color_primaries': 'bt709', 'color_range': 'tv', 'field_order': 'progres
sive', 'has_b_frames': 2, 'refs': 1, 'queue_size': 3, 'alert_total_dropped_segments': 18, 'process_cpu_pe
rcent': 55.0, 'process_memory_mb': 5564.75, 'process_threads': 23, 'spatialactivity': 52.52019010000001,
'temporalactivity': 11.01267696969697, 'blur': 4.112680000000001, 'blockiness': 0.9281734, 'blockloss': 1
.8257139999999998, 'freezing': 0.0, 'letterbox': 0.0, 'pillarbox': 0.0, 'blackout': 0.0, 'exposure': 132.
46, 'contrast': 49.9920111, 'interlace': 0.0006376999999999999, 'noise': 0.3502000000000001, 'flickering'
: 0.5964074999999999, 'volume_mean': -18.9, 'volume_max': -3.9, 'dropped_frames': 0, 'alert_dropped_fra
me': False, 'alert_blur': False, 'alert_blockiness': False, 'alert_blockloss': False, 'alert_freezing': Fa
lse, 'alert_uniform_frame': False, 'alert_black_frame': False, 'alert_no_audio': False, 'alert_silence': F
alse, 'count_alert_dropped_frames': 0, 'count_alert_blur': 1, 'count_alert_blockiness': 0, 'count_alert_
blockloss': 0, 'count_alert_freezing': 0, 'count_alert_uniform_frame': 0, 'count_alert_black_frame': 0, '
count_alert_no_audio': 0, 'count_alert_silence': 0, 'count_measurements': 88, 'mos_value': 3.22}
```

T5.3: Smart Media Distribution validation

Ongoing work

- Trial components deployment. [Media Production Control](#)

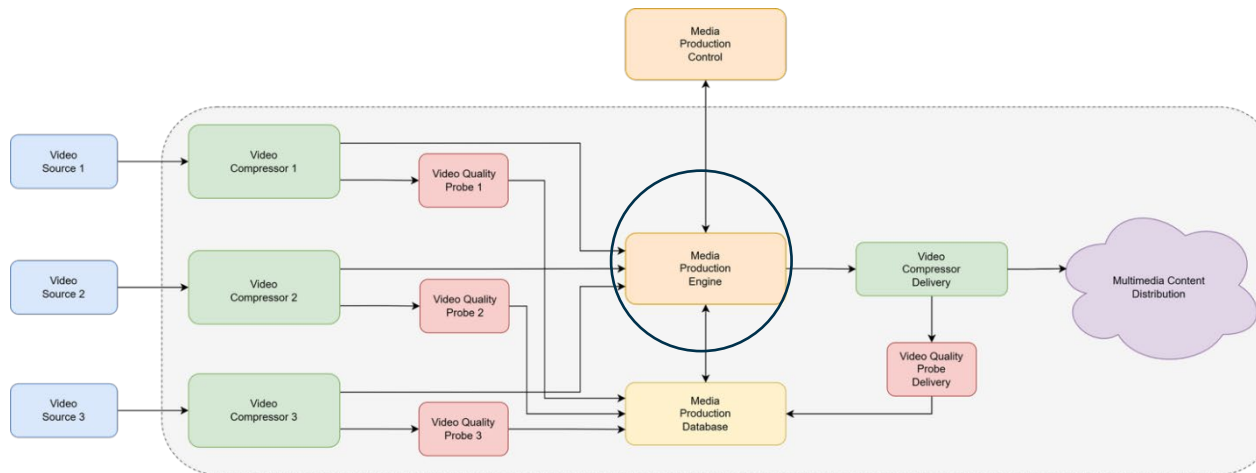




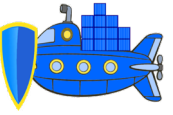
T5.3: Smart Media Distribution validation

Ongoing work

- Trial components deployment. [Media Production Engine](#)



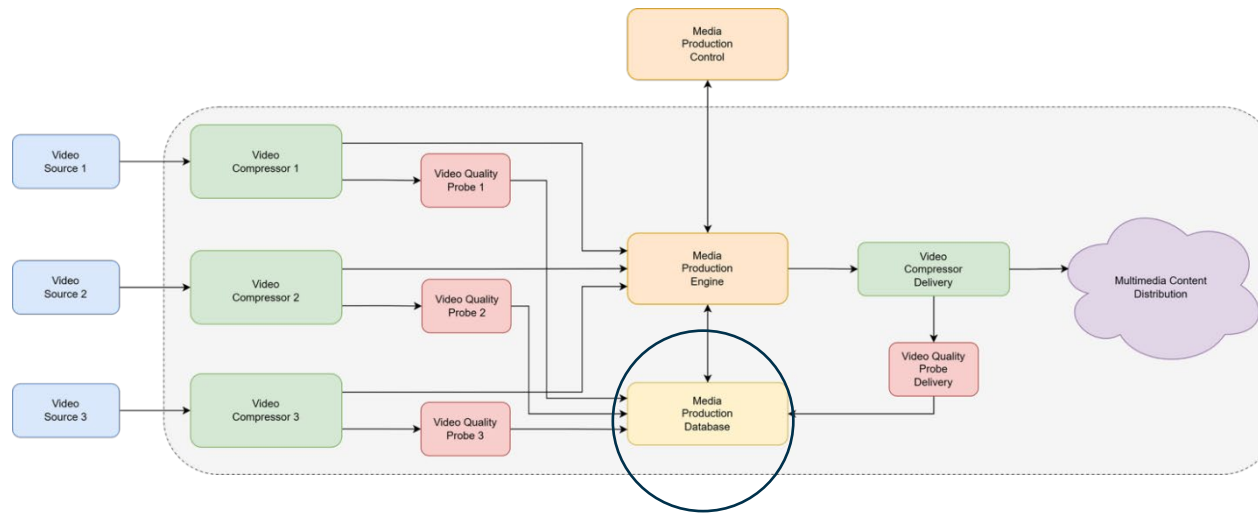
```
arya@arya-TEKNOPRO: ~/Desktop/Cybernemo/vproduction
arya@arya-TE... x arya@arya-TE... x arya@arya-TE... x arya@arya-TE... x
true, 'text1': 'CyberNEMO', 'text2': 'Trial 3 - Smart Media'}, 'audio_status': {'
cam1': 1.0, 'cam2': 0.0, 'cam3': 0.0, 'cam4': 0.0, 'break': 0.0, 'intro': 0.0},
'composite': 'fs(cam1,cam2)', 'stream_status': 'live', 'stream_info': {'video_re
solution': '1920x1080', 'video_fps': '25', 'video_codec': 'rawvideo', 'video_pix
_fmt': 'I420', 'calc_bandwidth_mbps_per_cam': 622.08, 'audio_sample_rate': '4800
0', 'audio_channels': '2', 'audio_fmt': 'S16LE'}, 'system_info': {'cpu_usage_per
cent': 58.1, 'ram_usage_percent': 36.5, 'ram_available_mb': 10077, 'uptime': '05
:34:08', 'session_duration': '00:04:57'}, 'network_info': {'core_ip': '192.168.0
.29'}, 'timestamp': '2026-04-10 16:32:01'}
2026-04-10 16:32:06 arya-TEKNOPRO __main__[15] INFO {'type': 'vproduction', 'vid
eo_status': {'channel_a': 'cam1', 'channel_b': 'cam2', 'logo1': True, 'logo2': T
true, 'text1': 'CyberNEMO', 'text2': 'Trial 3 - Smart Media'}, 'audio_status': {'
cam1': 1.0, 'cam2': 0.0, 'cam3': 0.0, 'cam4': 0.0, 'break': 0.0, 'intro': 0.0},
'composite': 'fs(cam1,cam2)', 'stream_status': 'live', 'stream_info': {'video_re
solution': '1920x1080', 'video_fps': '25', 'video_codec': 'rawvideo', 'video_pix
_fmt': 'I420', 'calc_bandwidth_mbps_per_cam': 622.08, 'audio_sample_rate': '4800
0', 'audio_channels': '2', 'audio_fmt': 'S16LE'}, 'system_info': {'cpu_usage_per
cent': 58.1, 'ram_usage_percent': 36.3, 'ram_available_mb': 10094, 'uptime': '05
:34:13', 'session_duration': '00:05:02'}, 'network_info': {'core_ip': '192.168.0
.29'}, 'timestamp': '2026-04-10 16:32:06'}
2026-04-10 16:32:11 arya-TEKNOPRO __main__[15] INFO {'type': 'vproduction', 'vid
eo_status': {'channel_a': 'cam1', 'channel_b': 'cam2', 'logo1': True, 'logo2': T
true, 'text1': 'CyberNEMO', 'text2': 'Trial 3 - Smart Media'}, 'audio_status': {'
cam1': 1.0, 'cam2': 0.0, 'cam3': 0.0, 'cam4': 0.0, 'break': 0.0, 'intro': 0.0},
'composite': 'fs(cam1,cam2)', 'stream_status': 'live', 'stream_info': {'video_re
solution': '1920x1080', 'video_fps': '25', 'video_codec': 'rawvideo', 'video_pix
_fmt': 'I420', 'calc_bandwidth_mbps_per_cam': 622.08, 'audio_sample_rate': '4800
0', 'audio_channels': '2', 'audio_fmt': 'S16LE'}, 'system_info': {'cpu_usage_per
cent': 56.8, 'ram_usage_percent': 36.3, 'ram_available_mb': 10104, 'uptime': '05
:34:18', 'session_duration': '00:05:07'}, 'network_info': {'core_ip': '192.168.0
.29'}, 'timestamp': '2026-04-10 16:32:11'}
```



T5.3: Smart Media Distribution validation

Ongoing work

- Trial components deployment. [Media Production Database](#)



The screenshot shows the RabbitMQ web interface at localhost:15672/#/queues. The 'Queues and Streams' tab is selected. The interface displays a table of queues with the following data:

Virtual host	Name	Type	Features	State	Ready	Unacked	Total	Incoming	deliver / get	ack
/	vcompressor-uc-media	classic	D	running	685	0	685	0.00/s		
/	vproduction-uc-media	classic	D	running	113	0	113	0.20/s	0.00/s	0.00/s
/	vqprobe-uc-media	classic	D	running	342	0	342	0.00/s		

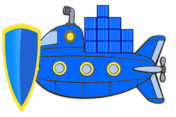
Below the table, there is a link to 'Add a new queue' and a footer with links to HTTP API, Documentation, Tutorials, New releases, Commercial edition, Commercial support, Discussions, Discord, and Plugins.



T5.3: Smart Media Distribution validation

CyberNEMO components in Trial 3

- ZTNA (Zero Trust Network Access)
- IPDM-DSS (Intrusion Prevention/Detection System)
- ZT-FML (Zero Trust Federated Machine Learning)
- CASB (Continuum Access Security Broker), the PRESS framework, and PPE (Privacy Protection Enforcement)
- SAAM (Knowledge Sharing, risk Assessment, threat Analysis and incidents Mitigation)



T5.3: Smart Media Distribution validation

KPIs

KPIs	Goal Statement	Target	To be discussed
1	Precision of identifying and mitigating cybersecurity threats	> 15%	Simulate attacks? What type? Who and how? Identification? DSS? Mitigation? Which CyberNEMO components?
2	Unauthorized access, ensuring content integrity in media production and distribution	< 20%	Simulate attacks? What type? Who and how? Unauthorized access? ZTNA (L2S-M)? Content integrity? ZTNA (PoT)? ZTNA (Notary)?
3	Treat/incidents detection/mitigation time, enhancing resilience and data loss protection	< 30%	Related to KPI 1 Mitigation? Which CyberNEMO components?
4	Latency via micro-segmentation, ensuring timely delivery during critical events	< 25%	Two tests, with and without micro-segmentation, to measure times and calculate latencies